

DEPARTMENT OF ELECTRICAL ENGINEERING

B.Tech. EE 8th Semester – Presentation Schedule for Project-II (EEEEEC23) End-Semester Evaluation

Presentation Details:

- **Date & Time:** 14th / 15th May 2026, starting at 10:00 AM (*final date shall be informed soon, start preparations for the dissertation printing*)
- **Venue:**
 - Section I: Room S-01 (Smart Block)
 - Section II: Room S-02 (Smart Block)

Presentation Guidelines:

- Each group will have approximately 10 minutes to present their work.
- The presence of all group members is mandatory.
- **A sample PowerPoint template (in black and white format) has already been shared via Google Classroom. Kindly ensure all text is in Times New Roman font with a WHITE BACKGROUND.**
- Attendance of all group members, including those from other sections or departments, is mandatory.
- PRESENTATIONS MUST BE PRELOADED
- **MOBILE PHONES ARE STRICTLY NOT ALLOWED.**

Important: No changes to the presentation slots or timings will be accommodated. Students arriving after 10:30 AM will not be allowed to present.

For any concerns, please contact:

Dr. V. S. K. V. Harish (Section I)
Dr. Ankit Kumar Singh (Section II)
BTP Coordinators

IN CASE OF ANY DISCREPANCY, immediately email the same with appropriate reason to: ankit@nsut.ac.in, vskv.harish@nsut.ac.in, and to: hod.ee@nsut.ac.in.

Report Guidelines

Three copies of the report (hardbound with Royal Blue cover; *see next page*) must be submitted at the time of presentation. The report must be as per the guidelines mentioned in this document. Students are required to present a draft copy of the thesis report to the concerned supervisor at **LEAST 10 DAYS (OR AS PER THE SUPERVISOR)** before the scheduled date of presentation for technical suggestions/review.

Guidelines for Project Report

1. The project report should be meticulously typed. Photocopied reports on ready-made material will not be entertained.
2. Plagiarism report signed by supervisors' should be attached at the end of the report. The similarity index should be less than 20 %.
3. **3-COPIES OF THESIS MUST BE HARD BOUND IN ROYAL BLUE COLOUR.** Each group is required to submit **two hard-bound copies** to the department; **one hard-bound copy** to each supervisor.
4. The format of the cover page is given separately. This also forms the first page of the project report.
5. Font should be Times Roman throughout the text. Font size for the text is 12pt and Heading 14pt. Line spacing should be 1.5
6. Margins (Left = 1.5", Right = 1", Top= 1", Bottom = 1").
7. **A new chapter should start with an odd page number (in case printing is done on both sides of the sheet).**
8. Equations should be written using Equation Editor (if MS Word is used). They should be centred and numbered properly, and numbering should be kept at the extreme right.
9. Give a number to each figure in the report. Place the caption just below it.
10. Give a list of figures at the beginning of the report.
11. Place Table captions just above the table. The captions should be numbered sequentially using Arabic numerals.
12. The references should be quoted in the text using square brackets. The references should be kept at the end of the report. The references should be listed in the order cited in the report. They should be written in the IEEE pattern.

Report Flow:

The report should have the following:

1. First page in the given format
2. Certificate [begin numbering pages with this page as number (i)] [signed by Supervisors]
3. Acknowledgement
4. Plagiarism Report (less than 20%, generated from an authenticated software)
5. Abstract (150 words) & keywords (max 6) [in one page]
6. List of contents (with page numbers)
7. List of figures with page numbers
8. List of tables with page numbers
9. Chapter 1 (Introduction)
10. Chapter 2 (Decided by the guide)
11. Chapter 3 (Decided by the guide)
12. Chapter 4 (Decided by the guide)
13. Chapter 5 (Conclusion)
14. References
15. Appendix

Sample Report

SOLAR ENERGY POWERED SMART HOME AUTOMATION USING IOT

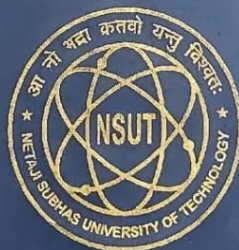
**Report submitted in partial fulfilment of the requirements
for the B.Tech. Degree in Electrical Engineering**

By

VAIBHAV KR. SHARMA	2020UEE4576
ABHIMANYU	2020UEE4584
AKSHIT KUMAR	2020UEE4600
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Under the supervision of

Dr. Alok Agrawal



**DEPARTMENT OF ELECTRICAL ENGINEERING
NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY
(NSUT) DWARKA, NEW DELHI**

May 2024

B.Tech. Project
On

POWER FACTOR IMPROVMENT

Report submitted in partial fulfillment of the requirements for the

B. Tech. degree in Electrical Engineering

By

Name of the Student

Roll No.

XXXX

XXXXXXXXXXXX

XXXXXXXXXX

XXXXXXXXXXXX

XXXXXX

XXXXXXXXXXXX

XXXXXXXXXXXXXX

XXXXXXXXXXXXXXXXXX

Under the supervision

of

(Dr. Harish Singh,

Dr. Suresh Kumar Singh)



DEPARTMENT OF ELECTRICAL ENGINEERING
NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY (NSUT) DWARKA,
NEW DELHI

MAY 2025

CANDIDATE(S) DECLARATION



DEPARTMENT OF ELECTRICAL ENGINEERING

I/We, Ankit (2018UEE3589), Arsh (2018UEE3592), Jhelum Awasthi (2018UEE3616) and Rahul Pandey (2018UEE3622) students of B. Tech., Department of Electrical Engineering, hereby declare that the Project-Thesis titled “Analysis and Optimization of Stiffened Panel Structures for Aerospace Applications Using Finite Element Simulation” which is submitted by me/us to the Department of Electrical Engineering, Netaji Subhas University of Technology (NSUT) Dwarka, New Delhi in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology is my/our original work and not copied from any source without proper citation. The manuscript has been subjected to plagiarism check by Turnitin software. This work has not previously formed the basis for the award of any other Degree.

Place:

Date:

Ankit
2018UEE3589

Arsh
2018UEE3591

Jhelum Awasthi
2018UEE3616

Rahul Pandey
2018UEE3622

CERTIFICATE OF DECLARATION



DEPARTMENT OF ELECTRICAL ENGINEERING

This is to certify that the work embodied in project thesis titled, **“Power Factor Improvement”** by Ankit (2018UEE3589), Arsh (2018UEE3592), Jhelum Awasthi (2018UEE3616) and Rahul Pandey (2018UEE3622) is the bonafide work of the group submitted to **Netaji Subhas University of Technology** for consideration in 8th Semester B.Tech. Project Evaluation.

The original Research work was carried out by the team under my/our guidance and supervision in the academic year 2024-2025. This work has not been submitted for any other diploma or degree of any university. On the basis of declaration made by the group, we recommend the project report for evaluation.

<FACULTY NAME>

(Designation)

Department of Electrical Engineering

Netaji Subhas University of Technology

ACKNOWLEDGEMENT

We would like to express my gratitude and appreciation to all those who make it possible to complete this project. Special thanks to our project supervisor(s) Dr. Ankit Kumar Singh and Dr. Alok Agrawal whose help, stimulating suggestions and encouragement helped us in writing this report. We also sincerely thank our colleagues for the time spent proofreading and correcting our mistakes.

We would also like to acknowledge with much appreciation the crucial role of the staff in the Electrical Engineering, who gave us a permission to use the lab equipment and also the machine and giving permission to use all necessary tools in the laboratory.

NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

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APPENDIX	

GROUPS TO APPEAR IN S-01 FOR B. TECH PROJECT – 2 MID SEMESTER EVALUATION

Group	Reporting Time	Group	Reporting Time
Grp #1 - #6	09.45 am	Grp #7 - #12	10.15 am
Grp #13 - #18	11.00 am	Grp #19 - onwards	11.15 am

Load your presentations on the venue computer before the reporting time.

Grp	Name of Supervisor	Title of Project	Roll No.	Name	Roll No.	Name	Roll No.	Name	Roll No.	Name
1.	Dr. Anjanee Kumar Mishra	Solar powered water pumping system using sensorless induction motor drive	2022UE E4533	Ishaan Mittal	2022UE E4528	Ayushi	2022UE E4544	Vishal Kumar		
2.	Dr. Astitva Kumar	Novel Metaheuristic based parameter estimation for PV Panels	2022UE E4513	Akshata Chopra	2022UE E1040	Yashika Gupta	2022UE E4052	Raima Rajan	2022UE E4522	Akshita Halsi
3.	Dr. V.S.K.V. Harish	Virtual Inertia Control	2022UE E6553	Vineet Goyal	2022UE E4518	Raman Singh	2022UE E4506	Aakarsh Gera	2022UE E4541	Kinshuk Upadhyaya
4.	Dr Subhodip Saha & Prof. (Dr) Subrata Mukhopadhyay	Scheduling of Coordinated Operation of Thermal Hydro & Renewable Energy Sources	2022UE E4581	Tanish Gaur	2022UE E4574	Arnav Gupta	2022UE E4588	Yuvraj Singh	2022UE E4555	Aryan
5.	Prof Prerna Gaur	Optimising of battery management system in EVs	2021UE E4549	Tanmoy Kant	2022UE E4590	Sahil Roy	2020UE E4532	Yash Pandey	NA	NA
6.	Dr. Astitva kumar	Physics based modelling for range estimation of electric vehicle	2022UE E4677	Priyam kumar	2022UE E4558	Komal verma	2022UE E4539	Ayushma n yadav	2022UE E4675	Mohd Imran
7.	Dr. Vineet kumar	Design of a bidirectional Dc Dc converter for battery charging and regeneration.	2022UE E4584	Razzak Ali	2022UE E4538	Aman	2022UE E4585	Alok kumar	2022UE E4536	Vaibhav kumar chaudhary
8.	Dr.Ravinder Singh	Intelligent coordination in decentralised multi-robots environment	2022UE E4577	Chetan Singh Pal	2022UE E4579	Pratik Kumar	2022UE E7655	Jharna Sharma	2022UE E4583	Hare Ram
9.	Dr. Subhodip Saha	Impact Assessment of Electric Vehicle Changing on distribution Network	2022UE E4626	Leena	2022UE E4578	Vinita Meena	2022UE E7625	Pragya Singh		
10.	Dr. Anjanee.K Mishra & Prof. (Dr) Subrata Mukhopadhyay	Impact of Renewable Energy On Power Transformer	2022UE E4556	Rohan Raj	2022UE E4551	Abhisar Kumar	Nikhil	2022UEE 4512	2022UE E6001	Ayush
11.	Dr. V. S. K. V. Harish	Peer To Peer Energy Sharing	2022UE E4582	Alen Shaju	2022UE E4546	Anshu Singh	2022UE E4564	Urishita Gupta	2022UE E4561	Aarav Kumar

GROUPS TO APPEAR IN S-01 FOR B. TECH PROJECT – 2 MID SEMESTER EVALUATION

Grp	Name of Supervisor	Title of Project	Roll No.	Name	Roll No.	Name	Roll No.	Name	Roll No.	Name
12.	Dr Ravinder Singh	Design a mesh based node architecture for UAV forensics	2022UE E4523	Jatin Sharma	2022UE E4501	Aditya Singh Yadav	2022UE E4534	Saurav Kumar	2022UE E4524	Prakhar mishra
13.	Dr. Ravinder Singh	Mitigation of dynamic and kinematic trajectory tracking error in autonomous mobile robots	2022UE E4509	Harshi Khandelwal	2022UE E4542	Siddharth Ahlawat	2022UE E4547	Suhani Malhotra	2022UE E4552	Vandit Sharma
14.	Dr. Ravinder Singh	Optimized Path Planning	2022UE E4540	Harsh Oswal	2022UE E3575	Krrishika Nim	2022UE E4096	Shivya Gupta	2022UE E4510	Vedant Shokeen
15.	Dr. Jyoti Yadav	Multimodal Analysis of Physiological Signals for Flow Recognition	2022UE E4543	Sanjoni Jain	2022UE E4568	Tiya Chadha	2022UE E4516	Vatsal Khosla		
16.	Dr.Astitva Kumar	Optimal Configuration of Bifacial PV Module	2022UE E4521	Ashish Upadhyay	2022UE E4586	Pratyai Chakrabarty	2022UE E4532	Aman	2022UE E4503	Aditya Garg
17.	Dr. Vineet kumar	Economic analysis and optimisation of off- grid Renewable energy systems	2022UE E4557	Rishab	2022UE E4566	Sanskar kumar	2022UE E4589	Aditya arya		
18.	Dr. Astitva Kumar	Performance simulation of BIPV Modules in Indian Context using PVSyst	2022UE E4535	Shaurya Singh	Tejas	2022UEE4 537	2022UE E4548	Arnav Khare	2022UE E4571	Shaz Ahmad
19.	Dr. Subhodeep Saha	Modelling of renewable energy management system using supercapacitor with battery	2022UE E4511	Jyotsana	2022UE E4567	Tushar Sagar				
20.	Dr.Vineet Kumar	Battery state estimation with imperfect data for Li-ion batteries	2022UE E4565	Dev Vats	Nimit Pant	2022UEE4 559	2022UE E4572	Vansh Jain	2022UE E4576	Varun Yadav
21.	Dr. Anjanee Kumar Mishra	Battery Management System for solar charging batteries	2022UE E4505	Udita Jarodia	2022UE E4570	Soumil Chola	2022UE E4554	Ayush Anand	2022UE E4560	Shreyansh Jindal
22.	Dr Ravinder Singh	Path optimization in dynamic environment	2022UE E4519	Kaviraj saini	2022UE E4527	Shagun sharma	2022UE E4526	Freya jain	2022UE E4573	Aashi gupta

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S.N o.	Roll No	Name of The Student	Group	Name of The Project	Name of The Supervisor	Mark s
1.	2022UEE4618	Ayush	1	Solar based EV charging system with AI/ML supported control techniques	Dr. Anjanee Kumar Mishra	
2.	2022UEE4622	Tushar				
3.	2022UEE4623	Yuvraj				
4.	2022UEE4629	Jayant				
5.	2022UEE4657	Shivam Tiwari	2	Optimal Scheduling of Electric Vehicle Charging	Dr. Subhodip Saha	
6.	2022UEE4664	Aman Aggarwal				
7.	2022UEE4666	Maryam				
8.	2022UEE4644	Kartik Bhandoria	3	Energy Management Strategy for a DC Microgrid including Electric Vehicle Fast Charging Stations	Dr. Vineet Kumar	
9.	2022UEE4673	Yash Tiwari				
10.	2022UEE4663	Pankaj Sharma				
11.	2022UEE4616	Gaurav Chauhan				
12.	2022UEE4183	Anshu Priya	4	Solar Powered Smart Irrigation System with Dump Load	Prof. Purna Gaur	
13.	2022UEE4613	Aayush Sharma				
14.	2022UEE4595	Krishna Gupta				
15.	2022UEE4599	Nikhil Sanwariya				
16.	2022UEE4597	Manish Pal	5	Battery Management system for Electric Vehicles	Dr. Ankit Kumar Singh	
17.	2022UEE4601	Mudit Jain				
18.	2022UEE4603	Priyanshu Mehra				
19.	2022UEE4632	Shubham Ray				
20.	2022UEE4606	Pawan Sharma	6	Green Hydrogen Production using solar energy	Dr. Ankit Kumar Singh	
21.	2022UEE4619	Md. Ashfaq				
22.	2022UEE4639	Harsh Pal				
23.	2022UEE4637	Abhinav Singh				
24.	2022UEE4635	Prashant Singh	7	V2G Energy Management System	Prof. Subrata Mukhopadhyay	
25.	2022UEE4655	Puneet Sahi				
26.	2022UEE4654	Prince Pal				
27.	2021UEE4619	Pratham Agarwal				
28.	2022UEE4641	Ansh Saran	8	Congestion mitigation in renewable rich power systems	Dr. Ankit Kumar Singh	
29.	2022UEE4656	Sahil sharma				
30.	2022UEE4592	Ayush joshi				
31.	2022UEE4645	Saurabh				
32.	2022UEE4615	Krishika verma	9	Optimal Real time Path routing for battery electric vehicles (BEVs)	Dr. Astitva Kumar	
33.	2022UEE4648	Ayush kumar				
34.	2022UEE4658	Aman mishra				
35.	2022UEE4636	Arun kumar				

36.	2022UEE4199	Mohit Anand	10	Novel Market Analysis Tool Using TOXO Smart Trainable Layer and Black Box LLM.	Dr. Astitva Kumar	
37.	2022UEE4190	Ankit Kumar				
38.	2022UEE4624	Vivek Kumar Upadhyay	11	Intelligent control strategy for frequency regulation in renewable integrated Power system	Dr. Deepak Kumar Gupta	
39.	2022UEE4659	Jatin				
40.	2022UEE4674	Dasham Saini				
41.	2022UEE4661	Rehan Akhtar				
42.	2022UEE4676	Aakash Jain	12	Snake Optimization Algorithm - A novel approach for blockchain scalability	Dr. Ankush Jain Dr. V.S.K.V Harish	
43.	2022UEE4612	Harshit Rai Verma				
44.	2022UEE4633	Ananya Gulati				
45.	2022UEE4621	Suchet	13	Fuel Cell based EVs: Integrating Green Hydrogen and Ethanol	Dr. Ankit Kumar Singh	
46.	2022UEE4640	Hitesh Kumar				
47.	2022UEE4549	Kirti				
48.	2022UEE4627	Chitranshu arya	14	Smart solar panel array with real time partial shading detection	Dr. Vineet Kumar	
49.	2022UEE4600	Kanishk jha				
50.	2022UEE4634	Shitiz solanki				
51.	2022UEE4617	Aayushi				
52.	2022UEE4598	Yash kanojia	15	Performance analysis of mppt algorithms for solar PV systems under partial shading conditions	Dr. Deepak Kumar Gupta	
53.	2022UEE4653	Dhruv Sharma				
54.	2022UEE4660	Utkarsh sharma				
55.	2022UEE4670	Tariq Aziz				
56.	2022UEE4665	Sahil Wadhawan	16	ML-Assisted Ant Colony Optimization of UPQC for THD and Cost Reduction	Dr. Subhodip Saha	
57.	2022UEE4668	Yashita Singh				
58.	2022UEE4669	Shreyansh Agarwal				
59.	2022UEE4631	Divya Kumari	17	Hybrid Mppt Techniques For Solar Pv Systems Under Partial Shading	Prof. Purna Gaur	
60.	2022UEE4642	Pallavi Kumari				
61.	2022UEE4651	Parth Keswani				
62.	2022UEE4662	Abhinav Singh				
63.	2022UEE4652	Nikhil Bajrang Kalaskar	18	AI-ML Based Battery Management System for EV's	Dr. Deepak Kumar Gupta	
64.	2022UEE4643	Rishabh Choudhary				
65.	2022UEE4650	Shivam Kumar Sahu				
66.	2022UEE4638	Simran Arora	19	Develop Bio Inspired Maximum Power Extraction method for pv systems	Dr. Astitva Kumar	
67.	2022UEE6605	Navya Pant				
68.	2022UEE3595	Lokendra				
69.	2022UEE4646	Aman Punia				
70.	2022UEE4607	Abhigya	20	Cell balancing techniques for	Dr. Alok Agrwal	

71.	2022UEE4647	Ekansh Yadav		Electric Vehicles		
72.	2022UEE4604	Prabhat Gupta				
73.	2022UEE4593	Chandra Rishabh Singh Nahariya				
74.	2022UEE4610	Sri Anshu Chaturvedi	21	Dynamic Wireless Charging System for EV using Arduino and Solar)	Dr. Prashant Kumar	
75.	2022UEE4630	Rohan Kumar				
76.	2022UEE4200	Rahul Kumar				
77.	2022UEE4620	Ghanshyam Chaudhary				
78.	2022UEE4611	Rahul Singh	22	Implementing FireCracker MicroVms For Renewable Energy Optimisation	Prof. Prerna Gaur	
79.	2022UEE4667	Vishakha Singh				
80.	2022UEE3641	Yashaswi	23	Forecasting of Bidding price in energy exchange using machine learning approach	Dr. Prashant Kumar	
81.	2022UEE4181	Ishika Agrahar				
82.	2022UEE3643	Pratham Pratap Bhalotia				
83.	2022EE4580	Aditya Singh				
84.	2022UEE4677	Priyam kumar	24	Physics based modelling for range estimation of electric vehicle	Dr. Astitva Kumar	
85.	2022UEE4675	Mohd Imran				
86.	2022UEE4558	Komal verma				
87.	2022UEE4539	Ayushman Yadav				
88.	2022UEE4628	Shivom Tyagi	25	Performance Analysis of Fuzzy MPPT Under Partial Shading	Prof. Prerna Gaur	
89.	2022UEE4596	Shivam Prasad				
90.	2022UEE4605	Rahul Kumar				
91.	2022UEE4671	Ayush				